

FoodSpy-Food inspection App

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This is a mobile app that helps users determine whether food is safe and healthy.
Users can:

1. Scan the QR code of the food
2. Check the source of the food (origin, processing, transportation)
3. Check the additives and health score
4. Receive food safety alerts (recall alerts)
5. Check the information of the food company

The system integrates multiple information sources to transform scattered food safety data into understandable information and actionable recommendations.

The Product: Food Inspection Application

Food health information mobile app is provides nearby food safety alerts news and personalized warnings if scanned foods are later linked to contamination issues users can also explore company profiles product origins and health ratings, Food safety news feeds recall databases user scan history geographic location data company transparency reports, Detect trends in food safety incidents identify regional risk patterns enable early alerts and improve consumer awareness through personalized information delivery, Data Science User Experience Information Management Product Management

The link: <https://www.fda.gov/food>.

The Wireframe

Information about how to check the report and related inspections content for the food company.

Static Data

- **Microbiological Safety**
- **Food Chemical Safety**
- Nutrition

Query

- The label of current food product

Info display

- The product has been approved by the app

Term	Meaning
product_id	Unique identifier for each food product
product_name	Name of the food item
producer_id	Identifier for the producer or manufacturer
origin_location	Geographic origin of the product
processing_steps	Description of processing and handling stages
ingredient_list	List of ingredients contained in the product
additive_list	List of additives used in the product
safety_rating	Safety level (e.g., Green / Yellow / Orange / Red)
certification	Food safety certifications (e.g., organic, FDA approved)
recall_status	Indicates whether the product has been recalled

Controlled Vocabulary

Derived / Calculated Data

Term	Meaning
health_score	Overall health score based on ingredients and additives
risk_level	Risk level derived from safety inspections and recalls
transparency_score	Degree of available and reliable product information

The structure of Basic Architecture System

Food information extract by Google Colab.

- **Batch Data:** Scheduled uploads of regulatory datasets (e.g., the **Food Inspection & Managment .csv** file, USDA FSIS bulk data).
- **Streaming/Real-time Data:** Live inputs from consumer mobile apps (e.g., the QR label scanner scanning a product in a grocery store).
- **External APIs:** Live connections to regional health alert feeds or food recall databases

Host: localhost

Information structure:

QR Code → **product_id**

Linked data:

- Product
- Producer
- Inspection
- Recall
- User Scan History

Data Flow

Scan → product_id → retrieve data → generate insights → display

Technologies

Google Colab for prototyping and data processing

Python for data integration, analysis, and alert logic

JSON for portable information structure and data exchange